Growth in Endogenous Production Networks Needs Compatible Recipes

A Correction to Acemoglu and Azar (2020)

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Abstract

Acemoglu and Azar (2020, *Econometrica*) show that when firms choose their suppliers from all available input combinations, the economy grows at rate $D\sum_{i,j}\beta_i\mathcal{L}_{ij}$: the growth rate D of the best attainable productivity, amplified by the Leontief inverse of the full potential input–output network (Theorem 6). Under the paper's stated assumptions this formula is false. In a simple economy that satisfies every assumption, the formula predicts growth 2 and equilibrium growth is 4/3. The best recipes require dropping an important supplier, and the assumptions control the best recipe but not the best recipe that keeps the important suppliers. The same happens with independent, continuous productivity draws. A second, separate failure comes from late-arriving industries that are very productive and gives growth 4. Both failures trace to specific steps in the proof. The exact growth formula is overturned in general, and with it Corollaries 1–3, the upper bound in Theorem 7, and the random-search contrast in Theorem B3. Sustained growth itself survives, as does the paper's benchmark with independent and identically distributed (iid) Gumbel or exponential draws. We state the condition, compatibility, under which the formula holds and show that it is also necessary.

1 The claim

At date t there are t goods. Industry i chooses a supplier set $S \subseteq \{1,\dots,t\}$, which gives log productivity $a_i(S)$ and Cobb–Douglas input shares α_{ij} for $j \in S$. With zero distortions and $q_i = -p_i$ (minus the log price), equilibrium solves

$$q_i \;=\; \max_{S\subseteq\{1,\dots,t\}} \Big\{ a_i(S) + \sum_{j\in S} \alpha_{ij}\, q_j \Big\}. \tag{1}$$

Each firm trades off its own productivity against cheaper inputs. With row sums below $\theta < 1$, (1) is a contraction with a unique solution. Nominal GDP is one, log real GDP is $\sum_i \beta_i(t) q_i$ with consumption shares $\beta_i(t) = \beta_i / \sum_{k\le t}\beta_k$, and growth is $g = \lim_t \frac{1}{t}\sum_i \beta_i(t) q_i$.

Assumption 4 (p. 54) requires, for each fixed industry i,

$$\frac{1}{t}\max_{S\subseteq\{1,\dots,t\}} a_i(S,t) \;\longrightarrow\; D > 0 \quad \text{almost surely.}$$

The paper notes that draws need not be identical or independent. Assumption 5 bounds row sums by $\theta < 1$ and makes the shares of late suppliers uniformly small. Theorem 6 (p. 55) concludes

$$\frac{-p_i(t)}{t} \;\longrightarrow\; D\sum_j \mathcal{L}_{ij}, \qquad g = D\sum_{i,j}\beta_i\mathcal{L}_{ij}, \qquad \mathcal{L} = (I-\alpha)^{-1}.$$

The paper's intuition (p. 56) is that each industry's productivity grows at rate D and the full network multiplies it. That requires firms to reach the productivity frontier *while also* using every important supplier. Nothing in the assumptions says they can.

2 Issue 1: the best recipes may exclude an important supplier

Example 1. *Let $\beta_i = 2^{-i}$ and give every industry the same shares, $\alpha_{i1} = \frac{1}{4}$ and $\alpha_{ij} = 2^{-j-1}$ for $j \ge 2$. The row sum is $\frac{1}{2}$, so $\sum_j \mathcal{L}_{ij} = 2$. Let*

$$a_i(S) = \begin{cases} \max S & \text{if } S \neq \emptyset \text{ and } 1 \notin S, \\ 0 & \text{if } 1 \in S \text{ or } S = \emptyset. \end{cases}$$

Recipes that use good 1 are less productive. Since $\max_S a_i(S) = t$, Assumption 4 holds with $D = 1$, uniformly in i. Assumptions 1′, 2′, 3 and 5 hold. Theorem 6 predicts $g = 1 \cdot 2 = 2$.

Equilibrium. Let $J_t = \{2, \dots, t\}$, $s_t = \sum_{j=2}^{t} \alpha_{ij} = \frac{1}{4}(1 - 2^{1-t})$ and $Q_t = t/(1 - s_t)$. Set $q_i = Q_t$ for all i. We check that no recipe beats J_t:

- A nonempty set without good 1 gives $\max S + Q_t \sum_{j \in S} \alpha_{ij} \le t + Q_t s_t = Q_t$, with equality only at J_t.
- A set containing good 1 gives at most $Q_t(s_t + \frac{1}{4})$, which is below $t + Q_t s_t$ because $Q_t/4 < t$.
- The empty set gives 0.

So Q_t solves (1), and by contraction it is the unique solution. Every firm drops good 1, and

$$\frac{\log Y_t}{t} = \frac{1}{1 - s_t} \;\longrightarrow\; \frac{1}{1 - 1/4} = \frac{4}{3} \;<\; 2.$$

At $t = 4, 8, 16, 32$ the ratio is 1.280, 1.330, 1.33332, 1.33333. The appendix code solves (1) by brute force over all 2^t recipes and gives the same numbers.

Why this is an issue. The economy satisfies the assumptions of the theorem and contradicts its conclusion, so the theorem is false as stated. The economics is that firms face a choice the formula ignores. Recipes at the frontier give rate $D = 1$ but forgo the supplier that carries half of all input shares, so the multiplier is only 4/3. Recipes that keep that supplier get the multiplier but no productivity growth. The formula combines the best productivity with the full network, but no firm can have both.

Not an artefact of a deterministic example. Let the $E_{i,S}$ be independent exponential draws with mean $1/\log 2$, one for every industry and recipe. Set $a_i(S) = E_{i,S}$ if $1 \notin S$ and $a_i(S) = c\, E_{i,S}$ if $1 \in S$, with $0 < c < 1$. Every draw is independent, continuous and positive; only the scale differs. The maximum of 2^{t-1} such draws grows like t, so Assumption 4 holds with $D = 1$, while recipes containing good 1 reach only rate c. Squeezing the equilibrium between the two kinds of recipe gives

$$g = \max\left\{\tfrac{4}{3},\; 2c\right\} < 2.$$

For $c < 2/3$ firms drop good 1. For $c > 2/3$ they keep it and use the full network, but their productivity grows at rate c, not D. With $c = 0.8$ the brute-force value at $t = 11$ is 1.5992, against a limit of 1.6. Holding the potential network and the frontier rate D fixed, growth can take any value in $[4/3, 2)$.

Where the proof slips. The proof (pp. 74–76) fixes a head $\{1, \dots, T\}$ that carries almost all input shares and takes the best recipe containing it, $S_i^{\max}(t)$. It then says $a_i(S_i^{\max}(t))$ "is the maximum of 2^{t-1-T} random variables" and that "Assumption 4 implies" it grows at rate D. With iid draws this is right, because the recipes that contain the head form a smaller copy of the full menu. Assumption 4, however, controls only the maximum over *all* recipes. In Example 1, recipes containing good 1 have productivity zero.

3 Issue 2: late industries that are very productive

The proof also moves from convergence for each fixed industry to convergence of $\min_{i \le t}$ and $\max_{i \le t}$, citing countable intersections. A countable intersection of almost-sure events gives one event of probability one. It does not give a common speed of convergence across a growing set of industries.

Example 2. *Keep the shares and β_i of Example 1, and let $a_i(\emptyset) = 0$ and $a_i(S) = 2^i + \max S$ otherwise. Adding suppliers never lowers productivity, so Issue 1 is absent. For each fixed i, $\max_S a_i(S)/t \to 1$, so Assumption 4 holds with $D = 1$.*

Every firm uses all goods. With $d_t = \sum_j \alpha_{ij} q_j$, the equilibrium is $q_i = t + 2^i + d_t$. Since $\sum_{j \le t} \alpha_{ij} 2^j = t/2$, we have $d_t/t \to 2$. So $q_i/t \to 3$ for every fixed industry, against the predicted 2, and

$$\frac{\log Y_t}{t} \;\longrightarrow\; 1 + 1 + 2 = 4,$$

where one unit comes from $\sum_{i \le t} \beta_i 2^i = t$. Brute force gives 3.999 at $t = 11$.

Why this is an issue. Each new industry's productivity level is large, but the assumption, stated industry by industry, allows it. These industries have tiny consumption shares, but they feed every other industry through the network, so they raise the growth of every fixed sector. The fix is a uniform bound, $\sup_t \max_{i \le t} |\max_S a_i(S,t)|/t < \infty$. Without it neither the formula nor its upper bound holds.

4 What gets overturned

Result	Status
Theorem 6 (growth formula)	False under the stated assumptions (Examples 1 and 2). Holds with compatibility and a uniform bound (Section 6).
Corollary 1 (finite set of flexible industries)	False. If only industry 1 chooses, with the technology of Example 1, and the others use labor only, growth is $\beta_1 = \frac{1}{2}$; the formula gives $\beta_1 \mathcal{L}_{11} = \frac{2}{3}$.
Corollary 2 ($n(t)$ new goods)	False; it includes $n(t) = t$, which is Example 1.
Corollary 3 (bounds D_1, D_2)	Lower bound false: Example 1 has $D_1 = D_2 = 1$ and growth $4/3 < 2$.
Theorem 7 (general technologies)	Upper bound $D/(1-\theta)$ false: Example 2 uses plain Cobb–Douglas and gives $q_i/t \to 3 > 2$.
Theorems 8 and 9 (essential inputs, creative destruction)	Inherit Issue 1, since they apply Theorem 6.
Theorem B3 (no growth without selection)	False under Assumption 4. With $a_i(S) = \max S$ and the shares of Example 1, a single random proposal already has productivity close to t, so growth is at least 1. Its proof treats Assumption 4 as if draws were iid.

The most important loss is interpretive. The formula splits growth into a technology rate D and a network multiplier $\sum_j \mathcal{L}_{ij}$ computed from the *potential* network, and the paper reads it that way ("direct and indirect effects", p. 56). The examples show that these two pieces cannot be measured separately and multiplied. Growth depends on which productivity rates are attainable together with which input shares.

5 What survives

- **Sustained growth.** If the frontier converges uniformly across industries, $\max_{i \le t} \left| \frac{1}{t} \max_S a_i(S,t) - D \right| \to 0$, then comparing (1) with constant vectors gives

$$D \;\le\; \liminf_t \frac{\log Y_t}{t} \;\le\; \limsup_t \frac{\log Y_t}{t} \;\le\; \frac{D}{1-\theta}.$$

  Selection among new input combinations still delivers positive long-run growth. What is lost is the exact rate, and with it the potential-network multiplier as a lower bound.

- **The paper's benchmark.** With iid Gumbel or exponential draws (Propositions B1–B2), compatibility holds, and so does the formula.

- **The static model.** Existence, uniqueness, efficiency and comparative statics (Theorems 1–5) do not use the growth argument and are not affected by these examples.

- **Empirical sections.** They are not assessed here.

6 The fix

For a finite set H of suppliers with $\alpha_{ij} > 0$, let $M_i^H(t) = \max_{H \subseteq S \subseteq \{1,\dots,t\}} a_i(S,t)$ be the best recipe that keeps H. Keep Assumptions 1′, 2′, 3 and the first part of 5, and add:

(A) *Compatibility:* for every i and every such finite H, $M_i^H(t)/t \to D$.

(B) *Uniform bound:* $\sup_t \max_{i \le t} |\max_S a_i(S,t)|/t < \infty$.

Proposition (corrected Theorem 6). *Under these assumptions, $q_i(t)/t \to r_i$ for every i, where $r = D(I-\alpha)^{-1}\mathbf{1}$, and $g = \sum_i \beta_i r_i$. Conversely, if (B) holds and $q_i(t)/t \to r_i$ for every i, then (A) holds.*

Sketch. By contraction and (B), $\|q(t)/t\|_\infty \le C$, so along a subsequence $q_i/t \to v_i$ for every i.

Lower bound. Fix K and use the best recipe that contains the positive-share suppliers $j \le K$. This gives $v_i \ge D + \sum_{j \le K} \alpha_{ij} v_j - C \sum_{j > K} \alpha_{ij}$. Letting $K \to \infty$ gives $v \ge D\mathbf{1} + \alpha v$, hence $v \ge r$.

Upper bound. We have $q_i \le \max_S a_i(S) + \sum_j \alpha_{ij} q_j^+$, and dominated convergence gives $v \le D\mathbf{1} + \alpha v$, hence $v \le r$. Every subsequence therefore has the same limit, and summable β gives g.

Necessity. At the chosen recipe, $a_i(S_i(t))/t = D + \sum_{j \notin S_i(t)} \alpha_{ij} r_j + o(1)$. The left side is at most $\max_S a_i(S)/t \to D$, and $r_j \ge D > 0$. So every positive-share supplier is eventually used and $a_i(S_i(t))/t \to D$, which squeezes $M_i^H(t)/t$ to D. □

Condition (A) says that insisting on any fixed set of important suppliers must not lower the long-run productivity rate. It holds with iid exponential or Gumbel draws, since the recipes containing H form a menu of $2^{t-|H|}$ iid draws. It also holds whenever adding suppliers never lowers productivity. It fails exactly when the best new technologies require abandoning established inputs. Assumption 5's tail condition is not needed.

7 Why it matters

Theorem 6 is the paper's main growth result, and its formula is the natural object to take to data: estimate a frontier rate, compute a Leontief inverse from input–output tables, multiply. The correction shows that this is valid only when frontier technologies are compatible with the existing supplier base. When they are not, and new methods require replacing important inputs, the economy gives up either productivity or network amplification, and growth falls below the potential-network prediction even though the frontier and the potential network are unchanged. Compatibility is a substantive economic restriction, not a technicality, and it is where the model's growth predictions should be tested.

Scope. Statements were checked against the published article and supplement. We found no erratum or comment raising these points, but our search was not exhaustive; corrections are welcome. We also found smaller problems elsewhere (Lemma 4 when conditioning on equilibrium prices with feedback loops, Theorem 10's outdegree lower bound, the displayed rate in Theorem 8, and Lemma A2), which we leave aside here.

A Code

Standard library only; it solves (1) by iteration over all 2^t recipes.

```
import itertools
def equilibrium(t, a):
    v = [0.25] + [2.0 ** -(j + 1) for j in range(2, t + 1)]
    sets = [S for r in range(t + 1) for S in itertools.combinations(range(1, t + 1), r)]
    q = [0.0] * t
    for _ in range(3000):
        new = [max(a(i, S) + sum(v[j-1] * q[j-1] for j in S) for S in sets)
               for i in range(1, t + 1)]
        done, q = max(abs(x - y) for x, y in zip(new, q)) < 1e-12, new
        if done: break
    beta = [2.0 ** -i for i in range(1, t + 1)]
    return sum(b * x for b, x in zip(beta, q)) / sum(beta) / t
A = lambda i, S: 0.0 if (not S or 1 in S) else max(S)
C = lambda c: (lambda i, S: 0.0 if not S else (c if 1 in S else 1.0) * max(S))
D = lambda i, S: 0.0 if not S else 2.0 ** i + max(S)
for t in (4, 8, 11):
    print(t, equilibrium(t, A), equilibrium(t, C(0.8)), equilibrium(t, D))
```

Output: $t = 4$: 1.2800, 1.5059, 3.8902; $t = 8$: 1.3299, 1.5938, 3.9922; $t = 11$: 1.3329, 1.5992, 3.9990. The limits are 4/3, 1.6 and 4; Theorem 6 predicts 2 in all three.

References

[1] Acemoglu, D., and P. D. Azar (2020). Endogenous Production Networks. *Econometrica* 88(1), 33–82, and its online supplement.